

Olympiad Test Paper — Mathematics (Class 7)

Chapter 14: Constructions and Tilings

Subject	Mathematics (NCERT Class 7)
Chapter	14 — Constructions and Tilings
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Constructions are described in words — picture the compass arcs and straightedge lines as you read. For tiling questions, the one rule that decides everything is that the angles meeting at a point must sum to **exactly 360°** ; watch the traps that a polygon tiles whenever 360° is "close", that you can trisect like you bisect, and that a construction is "just a careful drawing". Do not write on this paper — mark answers on your answer sheet.

1. In a construction, a "straightedge" is used to:

- (A) measure exact lengths in centimetres
- (B) draw straight lines only (it carries no markings)
- (C) measure angles like a protractor
- (D) swing arcs of equal radius

2. To construct the perpendicular bisector of a segment XY, you draw arcs from X and from Y. For the construction to work, those arcs must be drawn with:

- (A) any two different radii
- (B) a radius exactly equal to half of XY
- (C) a radius equal to the full length XY only
- (D) the same radius from both X and Y, that radius being more than half of XY

3. Every point lying on the perpendicular bisector of segment AB is:

- (A) equidistant from A and B
- (B) closer to A than to B
- (C) the midpoint of AB and nothing else
- (D) always inside segment AB

4. The perpendicular bisector construction crosses the segment XY at its midpoint AND meets it at an angle of:

- (A) 45°
- (B) 60°
- (C) 90°
- (D) 30°

5. Why does the equal-arc construction land exactly on the midpoint of XY, with no measuring?

- (A) Because the compass secretly measures half the length
- (B) Because each arc-crossing is the same distance from X as from Y, so the line through them is the segment's mirror line
- (C) Because any two arcs always cross at a midpoint
- (D) Because the straightedge is marked at its centre

6. A regular polygon's interior angle must satisfy which condition for the polygon to tile the plane by itself?

- (A) It must divide 360° a whole number of times
- (B) It must be a multiple of 90°
- (C) It must be less than 90°
- (D) It must be greater than 100°

7. Four squares meet exactly at a point in a square tiling because:

- (A) squares are the only four-sided shapes
- (B) squares have four equal sides
- (C) $4 \times 90^\circ = 360^\circ$, filling the full turn with no gap or overlap
- (D) two squares already make 360°

8. How many equilateral triangles meet at each vertex in a triangular tiling?

- (A) 3
- (B) 4
- (C) 5
- (D) 6

9. How many regular hexagons meet at each vertex when hexagons tile the plane?

- (A) 2
- (B) 3
- (C) 4
- (D) 6

10. A regular pentagon has interior angle 108° . Why can't regular pentagons tile the plane by themselves?

- (A) Pentagons have an odd number of sides
- (B) 108° is too large to fit any tiling
- (C) $360 \div 108$ is not a whole number, so they leave a gap or overlap at a vertex
- (D) Pentagons are not regular polygons

11. To construct a 60° angle with compass and straightedge, you essentially build the corner of which figure?

- (A) An equilateral triangle
- (B) A square
- (C) A regular hexagon
- (D) A right-angled isosceles triangle

12. Bisecting a 60° angle with only a compass and straightedge produces an angle of:

- (A) 15°
- (B) 45°
- (C) 20°
- (D) 30°

13. A student bisects a right angle, then bisects one of the two resulting angles. The smaller of the final pieces measures:

- (A) 45°
- (B) 30°
- (C) 22.5°
- (D) 15°

- 14.** Which of these can be constructed using only a compass and straightedge?
- (A) Trisecting a general 60° angle into three 20° angles
 (B) A 30° angle (by bisecting a constructed 60° angle)
 (C) A 20° angle directly (D) A 1° angle by repeated bisection
- 15.** The seven pieces of a tangram are cut from a single square and rearranged into the shape of a swan. The swan's total area compared with the original square is:
- (A) larger, because the swan looks bigger (B) smaller, because gaps appear
 (C) exactly the same (D) it depends on how the pieces are turned
- 16.** The deep reason rearranging tangram pieces conserves area is that:
- (A) sliding, turning or flipping a piece changes its position but not its size
 (B) the pieces are all triangles (C) a square always has the largest area
 (D) seven pieces always cover the same region
- 17.** A regular octagon has an interior angle of 135° . Can regular octagons tile the plane on their own?
- (A) Yes, four of them meet at a vertex (B) Yes, three of them meet at a vertex
 (C) Yes, two of them meet at a vertex
 (D) No, because $360 \div 135$ is not a whole number
- 18.** At a tiling vertex, two regular octagons (135° each) meet along with one more regular polygon that fits exactly. That polygon must have an interior angle of:
- (A) 45° (B) 90°
 (C) 120° (D) 60°
- 19.** The polygon in the previous situation (one that fills the leftover angle beside two octagons) is a:
- (A) regular hexagon (B) regular triangle
 (C) square (D) regular pentagon
- 20.** Three regular pentagons (108° each) are placed around a point. The angle still left uncovered is:
- (A) 36° (B) 0° (they tile)
 (C) 72° (D) 45°
- 21.** If your compass slipped so the two arcs for a perpendicular bisector had DIFFERENT radii, the line you draw would:
- (A) still be the exact perpendicular bisector
 (B) not pass through the midpoint and not be perpendicular — it would tilt and miss
 (C) become an angle bisector instead (D) be parallel to XY
- 22.** A "construction" in this chapter is special because the figure is exact by:

- (A) careful measuring with a ruler
- (B) tracing over a printed shape
- (C) using a protractor accurately
- (D) geometry — equal radii and equidistance — not measurement

23. Which statement about trisecting and bisecting angles is TRUE?

- (A) Bisecting a general angle is possible, but trisecting a general angle is impossible with these tools
- (B) Both bisecting and trisecting a general angle are easy with compass and straightedge
- (C) Trisecting is possible but bisecting is not
- (D) Neither can be done without a protractor

24. To get a 15° angle using only a compass and straightedge, the cleanest route is to:

- (A) construct 60° , bisect to 30° , bisect again to 15°
- (B) construct 45° and subtract 30°
- (C) trisect a 45° angle
- (D) it cannot be constructed at all

25. A regular dodecagon (12 sides) has interior angle 150° . Around a point you place one dodecagon corner. To fill the rest of the 360° with two MORE equal polygons, each of those must have interior angle:

- (A) 90°
- (B) 60°
- (C) 120°
- (D) 105°

26. Which set of regular polygons tile the plane EACH BY ITSELF?

- (A) Triangle, square, pentagon
- (B) Square, hexagon, octagon
- (C) Triangle, square, hexagon
- (D) Pentagon, hexagon, heptagon

27. A regular polygon has interior angle 144° (a regular decagon). The number of such polygons that would meet at a tiling vertex if they tiled would be $360 \div 144 = 2.5$. This tells you the decagon:

- (A) tiles, with 2 meeting at a vertex
- (B) cannot tile by itself, because 2.5 is not a whole number
- (C) tiles, with 3 meeting at a vertex
- (D) tiles, with 2.5 meeting at a vertex

28. Constructing the perpendicular bisector of segment XY gives you, as a free bonus, the:

- (A) exact midpoint of XY
- (B) angle bisector of angle X
- (C) a 60° angle at X
- (D) a tiling of the plane

29. A rectangle is cut along a diagonal into two triangles, which are then slid together into a different shape. The total area has:

- (A) doubled
- (B) halved
- (C) not changed
- (D) changed by the length of the diagonal

30. To bisect a given angle, from the vertex you draw an arc cutting both arms; then from those two crossing points you swing equal arcs to meet at a point P. The ray from the

vertex to P works because P is:

- (A) the midpoint of the original arc only
- (B) always 30° from each arm
- (C) the centre of the angle's circle
- (D) equidistant from both arms, so the ray splits the angle evenly

31. Equilateral triangles tile the plane. If instead you used the triangles to surround a point and only FIVE met there, the angle covered would be:

- (A) 360° (they still tile)
- (B) 300° , leaving a 60° gap
- (C) 240°
- (D) 320°

32. Which is the correct first step to construct a 60° angle at the end A of a ray AB?

- (A) From A, draw an arc crossing the ray; from that crossing, with the SAME radius, swing an arc cutting the first arc
- (B) Place the protractor at A and mark 60°
- (C) Measure 60 mm along the ray
- (D) Fold the paper at A

33. Squares tile the plane and so do regular hexagons. The reason hexagons need only THREE at a vertex while squares need FOUR is that:

- (A) hexagons are bigger than squares
- (B) hexagons have more sides so more must meet
- (C) a hexagon's interior angle (120°) is larger than a square's (90°), so fewer fill 360°
- (D) it is just a coincidence

34. A point is constructed at equal distance from both endpoints of a segment, on the same side. Joining it to a second such point on the other side produces a line that is the segment's:

- (A) angle bisector
- (B) perpendicular bisector
- (C) tangent
- (D) parallel

35. The interior angle of a regular polygon must divide 360° evenly for a self-tiling. The largest regular polygon (most sides) that can tile by itself is therefore the:

- (A) square
- (B) regular pentagon
- (C) regular octagon
- (D) regular hexagon

36. Tiling (tessellation) is best defined as:

- (A) covering a region with shapes leaving no gaps and no overlaps
- (B) covering a region with shapes that may overlap a little
- (C) drawing a shape with a compass
- (D) cutting a square into seven pieces

37. At a tiling vertex the corner angles meeting there must add to EXACTLY 360° . If they add to less than 360° , the result is:

- (A) an overlap of tiles
- (B) a perfect tiling
- (C) a gap (uncovered wedge)
- (D) a 90° corner

- 38.** If the angles meeting at a tiling vertex add to MORE than 360° , the result is:
 (A) a gap (B) an overlap (tiles pile on top of each other)
 (C) a perfect fit (D) a perpendicular bisector
- 39.** A mixed tiling uses squares (90°) and equilateral triangles (60°). At one vertex three triangles and two squares meet. The angles there total:
 (A) 330° (B) 300°
 (C) 390° (D) 360°
- 40.** A mixed tiling uses regular hexagons (120°) and equilateral triangles (60°). At a vertex two hexagons already meet; the number of triangles needed to complete the full 360° is:
 (A) two triangles (B) one triangle
 (C) three triangles (D) four triangles
- 41.** A 90° angle is constructed and then bisected twice in succession (each time taking the smaller new angle and halving it). After the two bisections, the smallest angle present measures:
 (A) 45° (B) 30°
 (C) 22.5° (D) 60°
- 42.** Which of these angles is NOT directly constructible by combining a constructed 60° angle with bisections of it?
 (A) 30° (B) 15°
 (C) 90° (as $60^\circ + 30^\circ$) (D) 20°
- 43.** You construct a 60° angle, then bisect it, then place the 30° angle next to the original 60° . The total angle formed is:
 (A) 45° (B) 30°
 (C) 120° (D) 90°
- 44.** A tangram square is rearranged into a tall thin "tree" and into a wide squat "boat", using all seven pieces each time. The tree and the boat have:
 (A) equal areas (B) different areas, the boat larger
 (C) different areas, the tree larger (D) areas that cannot be compared
- 45.** Which statement is the correct reason only triangles, squares and hexagons tile among regular polygons?
 (A) They are the only shapes with equal sides
 (B) Only their interior angles (60° , 90° , 120°) divide 360° a whole number of times
 (C) They are the only convex polygons
 (D) Larger polygons are simply too heavy to tile
- 46.** A regular polygon whose interior angle is 120° will, at a tiling vertex, have how many of its corners meeting?

- (A) 2
(C) 3
- (B) 4
(D) 6

47. The interior angle of an equilateral triangle is 60° . The number meeting at a tiling vertex (six) multiplied by 60° gives:

- (A) 360°
(C) 420°
- (B) 300°
(D) 240°

48. You want to construct a 45° angle with compass and straightedge. The correct route is to:

- (A) trisect a 135° angle
(C) bisect a 60° angle
- (B) measure it with a protractor
(D) construct a right angle (90°) and bisect it

49. The perpendicular bisector of a segment is, in precise terms, the set of ALL points that are:

- (A) inside the segment
(C) closer to the left endpoint
- (B) the same distance from both endpoints
(D) exactly the midpoint

50. Three regular hexagons (120° each) meet at a vertex. The angle they cover is:

- (A) 240°
(C) 360°
- (B) 300°
(D) 480°

51. A student claims "any regular polygon can tile the plane." The simplest single counterexample is the:

- (A) square
(C) equilateral triangle
- (B) regular pentagon
(D) regular hexagon

52. To construct a 90° angle (a perpendicular at a point P on a line) the construction relies on building a:

- (A) 60° angle and doubling it
(C) tangram
- (B) regular pentagon
(D) perpendicular bisector / equal-arc construction so the line meets at a right angle

53. Two equilateral triangles and two squares are arranged around a point. The total of their corner angles is:

- (A) 300°
(C) 330°
- (B) 360°
(D) 270°

54. A regular polygon's interior angle is 140° . The value $360 \div 140$ is between 2 and 3 (about 2.57). This polygon:

- (A) tiles with 2 meeting at a vertex
(C) cannot tile the plane by itself
- (B) tiles with 3 meeting at a vertex
(D) tiles with 2.57 meeting at a vertex

55. Bisecting an angle requires finding a point equidistant from both arms. The tool that locates such a point exactly, without measuring, is:

- (A) a protractor (B) equal-radius compass arcs
(C) a marked ruler (D) a set square

56. A floor is to be tiled with regular pentagons only. Pushing three pentagons (108° each) snug at a corner leaves a wedge. Four pentagons would overshoot. This shows pentagons:

- (A) tile with three at a vertex (B) tile with four at a vertex
(C) tile only if cut in half
(D) cannot tile, since neither 3 nor 4 of them makes exactly 360°

57. Which pair of facts is BOTH true?

- (A) A general angle can be bisected with compass and straightedge; pentagons cannot tile the plane
(B) A general angle can be trisected with compass and straightedge; pentagons tile the plane
(C) Constructions need a protractor; hexagons cannot tile
(D) The perpendicular bisector misses the midpoint; squares cannot tile

58. The interior angle of a regular polygon increases as the number of sides increases. This is why, among regular polygons, the ability to self-tile STOPS after the hexagon: any polygon with more sides has an interior angle that is:

- (A) less than 90° , too small
(B) more than 120° but not a whole-number divisor of 360°
(C) exactly 360° (D) negative

59. A 30° angle has been constructed (by bisecting 60°). Constructing one more 30° angle beside it and then a third gives a total angle of:

- (A) 60° (B) 120°
(C) 90° (D) 30°

60. The single rule that governs ALL of plane tiling — deciding which shapes fit around a point — is that the angles meeting at any vertex must sum to:

- (A) exactly 180° (B) exactly 270°
(C) any amount up to 360° (D) exactly 360°

Solutions — Mathematics (Class 7), Chapter 14: Constructions and Tilings

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	A	21	B	31	B	41	C	51	B
2	D	12	D	22	D	32	A	42	D	52	D
3	A	13	C	23	A	33	C	43	D	53	A
4	C	14	B	24	A	34	B	44	A	54	C
5	B	15	C	25	D	35	D	45	B	55	B
6	A	16	A	26	C	36	A	46	C	56	D
7	C	17	D	27	B	37	C	47	A	57	A
8	D	18	B	28	A	38	B	48	D	58	B
9	B	19	C	29	C	39	D	49	B	59	C
10	C	20	A	30	D	40	A	50	C	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

- (B)** A straightedge is an *unmarked* ruler — its only job is to draw straight lines. It carries no scale (so it cannot measure lengths) and is not a protractor; swinging arcs is the compass's job.
- (D)** Both arcs must be the SAME radius (so each crossing is equidistant from X and Y), and that radius must exceed half of XY or the arcs won't reach far enough to cross. "Exactly half" makes the arcs meet only on the segment (no upper/lower crossings); different radii destroy the equidistance.
- (A)** The perpendicular bisector is precisely the locus of points equidistant from A and B. The midpoint is just ONE such point; the whole line is equidistant, and points on it can lie far outside the segment.
- (C)** By definition the perpendicular bisector meets the segment at a right angle, 90° , while also cutting it in half. That is exactly the "perpendicular" in the name.
- (B)** Each arc-crossing is found at equal radius from X and from Y, so it is equidistant from both endpoints — it lies on the segment's mirror line. Two such crossings fix that

mirror line, which cuts XY at its midpoint. No length is ever measured; equidistance does the work.

6. (A) For corners to surround a point with no gap/overlap, copies of the interior angle must fill exactly 360° , i.e. $360^\circ \div (\text{interior angle})$ must be a whole number. Being a multiple of 90° or "small" is neither necessary nor sufficient (120° tiles though it isn't a multiple of 90°).

7. (C) A square's interior angle is 90° , and four of them give $4 \times 90^\circ = 360^\circ$ — a full turn with no gap or overlap. Having four equal sides (B) is true but is not the tiling reason; two squares give only 180° .

8. (D) Each interior angle of an equilateral triangle is 60° , and $360^\circ \div 60^\circ = 6$, so six triangles meet at every vertex.

9. (B) A regular hexagon's interior angle is 120° , and $360^\circ \div 120^\circ = 3$, so three hexagons meet at each vertex (the honeycomb pattern).

10. (C) A regular pentagon's interior angle is 108° , and $360 \div 108 \approx 3.33$ — not a whole number. Three pentagons give 324° (a 36° gap) and four give 432° (overlap), so they cannot surround a point cleanly.

11. (A) The 60° construction copies the corner of an equilateral triangle: equal-radius arcs build three equal sides, whose corners are all 60° . A square's corner is 90° , a hexagon's 120° .

12. (D) Bisecting halves the angle: $60^\circ \div 2 = 30^\circ$. (Bisecting again would give 15° , the trap option A.)

13. (C) Bisect $90^\circ \rightarrow$ two 45° angles. Bisect one $45^\circ \rightarrow$ two 22.5° pieces. The smallest piece present is 22.5° . (45° is the in-between stage; 30° comes from bisecting 60° , not 90° .)

14. (B) A 30° angle is constructible — bisect a constructed 60° . Trisecting a general angle (so $60^\circ \rightarrow$ three 20° s) is provably impossible with these tools, so a 20° angle "directly" is out, and an exact 1° cannot be reached by halving from $60^\circ/90^\circ$ either.

15. (C) Rearranging the same seven pieces only moves them; nothing is added or removed, so the swan has exactly the same area as the original square. "Looks bigger" and "gaps appear" are perception traps.

16. (A) Sliding, turning or flipping a piece is a rigid motion: it changes position, never size. So a fixed set of pieces re-adds to the same total area no matter the arrangement.

17. (D) A regular octagon's interior angle is 135° , and $360 \div 135 \approx 2.67$ — not a whole number. Two octagons give 270° (a 90° gap), three give 405° (overlap), so octagons cannot tile alone.

- 18. (B)** Two octagons contribute $2 \times 135^\circ = 270^\circ$. The leftover to reach a full 360° is $360^\circ - 270^\circ = 90^\circ$, so the third polygon's interior angle is 90° .
- 19. (C)** A regular polygon with a 90° interior angle is a square. (This is the classic octagon–square semi-regular tiling.)
- 20. (A)** Three pentagons give $3 \times 108^\circ = 324^\circ$. The angle left uncovered is $360^\circ - 324^\circ = 36^\circ$. (A 0° leftover would mean they tile, which they don't.)
- 21. (B)** Unequal radii destroy the equidistance: the two crossings are no longer the same distance from X and Y, so the line through them tilts, misses the midpoint, and is not perpendicular. It is neither the true perpendicular bisector nor an angle bisector.
- 22. (D)** A construction is *exact by geometry* — equal radii force equal distances, guaranteeing the result without any measuring. Ruler/protractor measuring (A, C) or tracing (B) would only be approximate; that is precisely what makes constructions special.
- 23. (A)** With compass and straightedge any angle can be BISECTED, but trisecting a *general* angle is provably impossible (you can make $30^\circ = \text{half of } 60^\circ$, but not $20^\circ = \text{a third}$). So only A is true.
- 24. (A)** Construct 60° (equilateral corner), bisect to 30° , bisect again to 15° . Subtracting 30° from 45° isn't a single clean construction step here, and trisecting a 45° is impossible; 15° certainly IS constructible, ruling out D.
- 25. (D)** One dodecagon corner is 150° . The remaining angle is $360^\circ - 150^\circ = 210^\circ$, shared by two equal polygons: $210^\circ \div 2 = 105^\circ$ each.
- 26. (C)** Only triangle (60°), square (90°) and hexagon (120°) have interior angles dividing 360° a whole number of times, so only this trio tiles each on its own. Pentagons (108°) and octagons (135°) do not.
- 27. (B)** $360 \div 144 = 2.5$ is not a whole number, so a whole number of decagons can never surround a point — the decagon cannot tile by itself. "2.5 tiles meet" (D) is meaningless; you can't have half a tile-corner at a vertex.
- 28. (A)** The perpendicular-bisector construction cuts XY exactly in half, so it hands you the midpoint for free — no measuring. It does not produce an angle bisector at X, a 60° angle, or a tiling.
- 29. (C)** Cutting and rearranging is rigid motion only — the two triangles are merely slid into a new outline, so the total area is unchanged (conservation of area under rearrangement). The diagonal's length has nothing to do with total area.
- 30. (D)** P is built at equal distances from the two crossing points on the arms, which makes it equidistant from both arms. A point equidistant from both arms lies on the

angle's bisector, so the ray $V \rightarrow P$ splits the angle into two equal halves.

31. (B) Five equilateral triangles give $5 \times 60^\circ = 300^\circ$, which is 60° short of a full 360° , leaving a 60° gap. (Six are needed to actually tile.)

32. (A) With compass and straightedge you start by swinging an arc from A across the ray, then from that crossing point swing the SAME radius to cut the arc — building the equilateral corner. Protractors, measuring or folding are not construction steps.

33. (C) The count is $360^\circ \div (\text{interior angle})$. A hexagon's 120° is larger than a square's 90° , so fewer hexagons fill the 360° (three vs four). It has nothing to do with physical size or "more sides means more meet".

34. (B) Two points each equidistant from both endpoints (one on each side) lie on the segment's mirror line; joining them gives the perpendicular bisector. Equidistance from the *endpoints* (not from arms) is the perpendicular-bisector signature.

35. (D) Self-tiling needs the interior angle to divide 360° evenly. Among regular polygons that holds only for the triangle, square and hexagon; the hexagon has the most sides of these, so it is the largest self-tiler. The octagon (135°) does not divide 360° .

36. (A) Tiling (tessellation) means covering a region with shapes leaving NO gaps and NO overlaps. "May overlap a little" (B) violates the definition; the others describe constructions/tangrams.

37. (C) Less than 360° at a vertex means the corners don't reach all the way round — an uncovered wedge, i.e. a gap. (More than 360° would force an overlap.)

38. (B) More than 360° means the corners run past a full turn and pile on top of one another — an overlap. (Less than 360° gives a gap; exactly 360° is the perfect fit.)

39. (D) Three triangles + two squares = $3 \times 60^\circ + 2 \times 90^\circ = 180^\circ + 180^\circ = 360^\circ$. The angles fill a full turn exactly, so this is a valid vertex.

40. (A) Two hexagons contribute $2 \times 120^\circ = 240^\circ$, leaving $360^\circ - 240^\circ = 120^\circ$ to fill. Equilateral triangles are 60° each, so $120^\circ \div 60^\circ = 2$ triangles complete the vertex. One triangle gives only 300° (a gap); three give 420° and four give 480° (overlaps).

41. (C) Bisect $90^\circ \rightarrow 45^\circ$; bisect $45^\circ \rightarrow 22.5^\circ$. After two bisections the smallest angle present is 22.5° . (45° and 30° are intermediate/distractor values.)

42. (D) From a constructed 60° you can bisect to 30° , bisect again to 15° , and add to make $90^\circ (= 60^\circ + 30^\circ)$. But 20° would require trisecting 60° , which is impossible — so 20° is NOT constructible this way.

43. (D) The original is 60° ; bisecting gives a 30° angle; placing the 30° beside the 60° gives $60^\circ + 30^\circ = 90^\circ$. (120° would be $60^\circ + 60^\circ$.)

44. (A) Both the "tree" and the "boat" use all seven pieces of the same square, so by conservation of area under rearrangement they have exactly equal areas. Appearance (tall vs wide) does not change total area.

45. (B) The reason is pure arithmetic: only 60° , 90° and 120° divide 360° a whole number of times (6, 4 and 3 times). Equal sides, convexity and "weight" are irrelevant to whether a regular polygon tiles.

46. (C) $360^\circ \div 120^\circ = 3$, so three of its corners meet at a vertex (this is the hexagon case).

47. (A) $6 \times 60^\circ = 360^\circ$, a full turn — which is exactly why six equilateral triangles tile around a point.

48. (D) Construct a right angle (90°) and bisect it: $90^\circ \div 2 = 45^\circ$. Bisecting 60° gives 30° , not 45° ; trisecting 135° is impossible; a protractor isn't a construction.

49. (B) Precisely, the perpendicular bisector is the set of ALL points the same distance from both endpoints (the locus). The midpoint (D) is just one of those points, not the whole set.

50. (C) $3 \times 120^\circ = 360^\circ$, a full turn — so three hexagons exactly surround a vertex with no gap or overlap.

51. (B) The regular pentagon (108°) is the simplest counterexample: 108° does not divide 360° , so it cannot tile alone. Triangle, square and hexagon all DO tile, so they're not counterexamples.

52. (D) A right angle is built from the equal-arc / perpendicular construction, which by design meets the line at 90° . Doubling a 60° gives 120° (not 90°); a pentagon or tangram is irrelevant.

53. (A) $2 \times 60^\circ + 2 \times 90^\circ = 120^\circ + 180^\circ = 300^\circ$. That is 60° short of 360° , so it is NOT a valid vertex — but the question only asks the total, which is 300° .

54. (C) $360 \div 140 \approx 2.57$ is not a whole number, so no whole number of these polygons can surround a point — it cannot tile by itself. ("2.57 meeting" is meaningless.)

55. (B) Equal-radius compass arcs locate a point equidistant from both arms exactly, without measuring — that is what makes angle bisection a true construction. A protractor or marked ruler would only measure approximately.

56. (D) Three pentagons give 324° (a 36° wedge left) and four give 432° (overlap); since neither hits exactly 360° , regular pentagons cannot tile the plane. Cutting them in half

changes the shape, so that doesn't count.

57. (A) True facts: a general angle CAN be bisected with these tools, and regular pentagons CANNOT tile. Option B reverses both (trisection is impossible; pentagons don't tile), and C, D each contain a false claim (constructions need no protractor; hexagons DO tile; the perpendicular bisector hits the midpoint; squares DO tile).

58. (B) Beyond the hexagon, each regular polygon's interior angle exceeds 120° (heptagon $\approx 128.6^\circ$, octagon 135° , ...) yet never again divides 360° a whole number of times, so none of them can self-tile. The angle is never less than 90° , never 360° , never negative.

59. (C) Three 30° angles placed in a row give $3 \times 30^\circ = 90^\circ$. (Two would give 60° ; four would give 120° .)

60. (D) The single governing rule of plane tiling is that the corner angles meeting at any vertex must sum to EXACTLY 360° — a full turn — with no gap (less) and no overlap (more). "Any amount up to 360° " is the trap that allows gaps.

Solutions complete: 60 / 60.